Class 08: Breast Cancer Mini Project

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a data-set from fine needle aspiration (FNA) of a breast mass.

Data Import

The data is made available as a CSV file for download. We can read this in using `read.csv()`:

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names = 1)
```

```
head(wisc.df, 3)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38          122.8      1001
842517           M       20.57        17.77          132.9      1326
84300903         M       19.69        21.25          130.0      1203
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
         perimeter_worst area_worst smoothness_worst compactness_worst
842302             184.6       2019           0.1622            0.6656
842517             158.8       1956           0.1238            0.1866
84300903           152.5       1709           0.1444            0.4245
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
```

Make sure we remove or exclude the `diagnosis` column from the data-set that we use for further analysis - this is the expert diagnosis as either M or B.

```r
# We can use -1 here to remove the first column
wisc.data <- wisc.df[,-1]
diagnosis <- as.factor( wisc.df$diagnosis )
```

Exploratory data analysis

> Q1. How many observations are in this dataset?

```r
nrow(wisc.data)
```

```
[1] 569
```

There are 569 observations.

> Q2. How many of the observations have a malignant diagnosis?

```r
table(wisc.df$diagnosis)
```

```

  B   M 
357 212 
```

212 observations have a malignant diagnosis.

> Q3. How many variables/features in the data are suffixed with _mean?

We can use the `grep()` function to help us here:

```r
colnames(wisc.data)
```

```
 [1] "radius_mean"             "texture_mean"           
 [3] "perimeter_mean"          "area_mean"              
 [5] "smoothness_mean"         "compactness_mean"       
 [7] "concavity_mean"          "concave.points_mean"    
 [9] "symmetry_mean"           "fractal_dimension_mean" 
[11] "radius_se"               "texture_se"             
[13] "perimeter_se"            "area_se"                
[15] "smoothness_se"           "compactness_se"         
[17] "concavity_se"            "concave.points_se"      
[19] "symmetry_se"             "fractal_dimension_se"   
[21] "radius_worst"            "texture_worst"          
[23] "perimeter_worst"         "area_worst"             
[25] "smoothness_worst"        "compactness_worst"      
[27] "concavity_worst"         "concave.points_worst"   
[29] "symmetry_worst"          "fractal_dimension_worst"
```

```r
length(grep("_mean", colnames(wisc.data), value = TRUE))
```

```
[1] 10
```

There are 10 variables in the data suffixed with _mean.

Principal Component Analysis (PCA)

We need to scale our data before PCA with the `scale-TRUE` argument to `prcomp()`.

```r
wisc.pr <- prcomp(wisc.data, scale = TRUE)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

> Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

44.27% is captured by the first principal component.

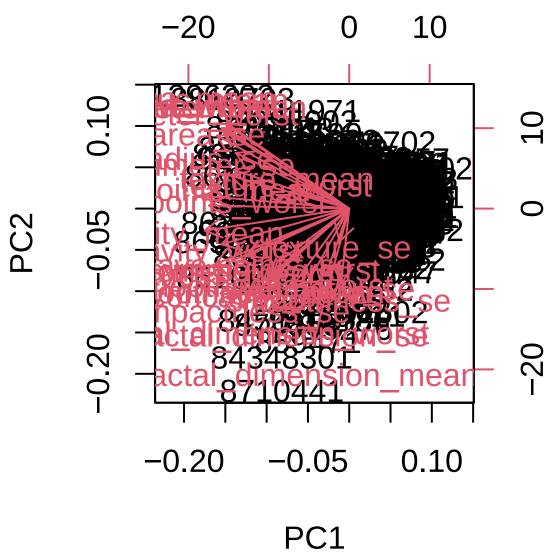
> Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

The first three principal components are required to describe at least 70% of the original variance in the data.

> Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

The first seven principal components are required to describe at least 90% of the original variance in the data.

```r
biplot(wisc.pr)
```

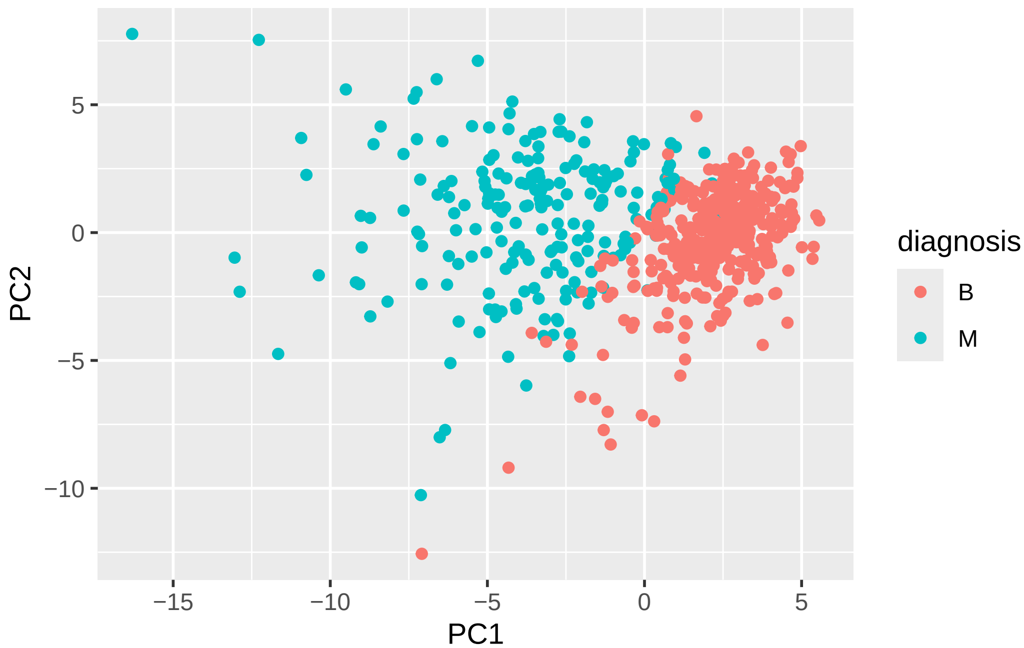


> Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

This plot is incredibly difficult to understand as conclusions cannot be made from the plot with everything crowded together.

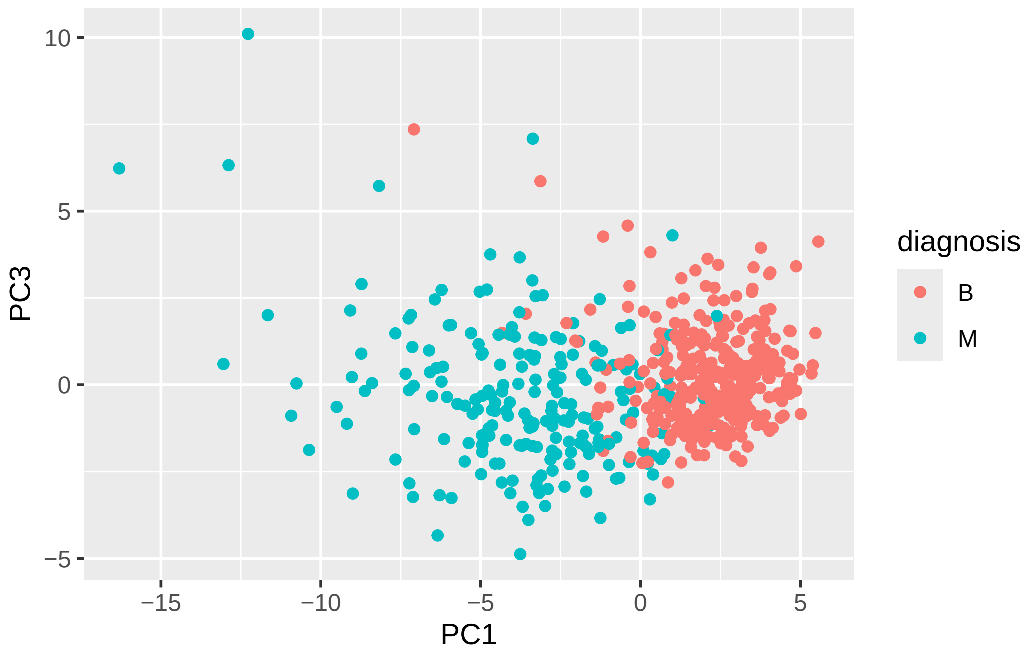
```r
library(ggplot2)

ggplot(wisc.pr$x) +
```

```
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



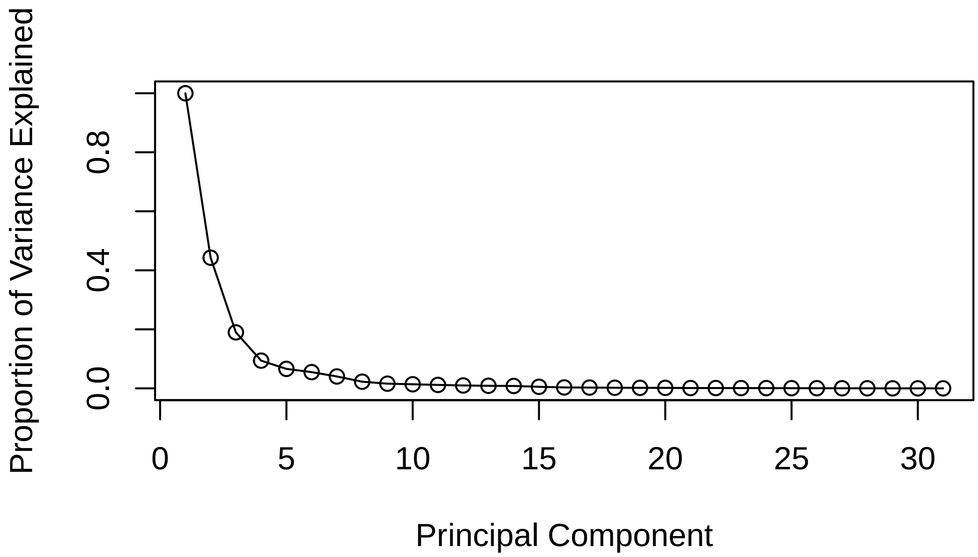
For both plots, the malignant observations are towards the left of the graph whereas the benign are assembled towards the right. This presents a separation between the benign and malignant samples.

Variance explained

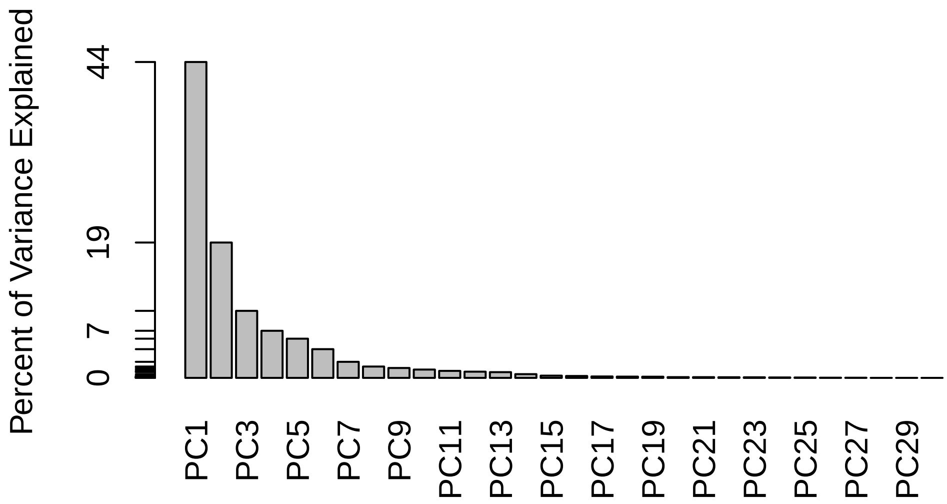
```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```




```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



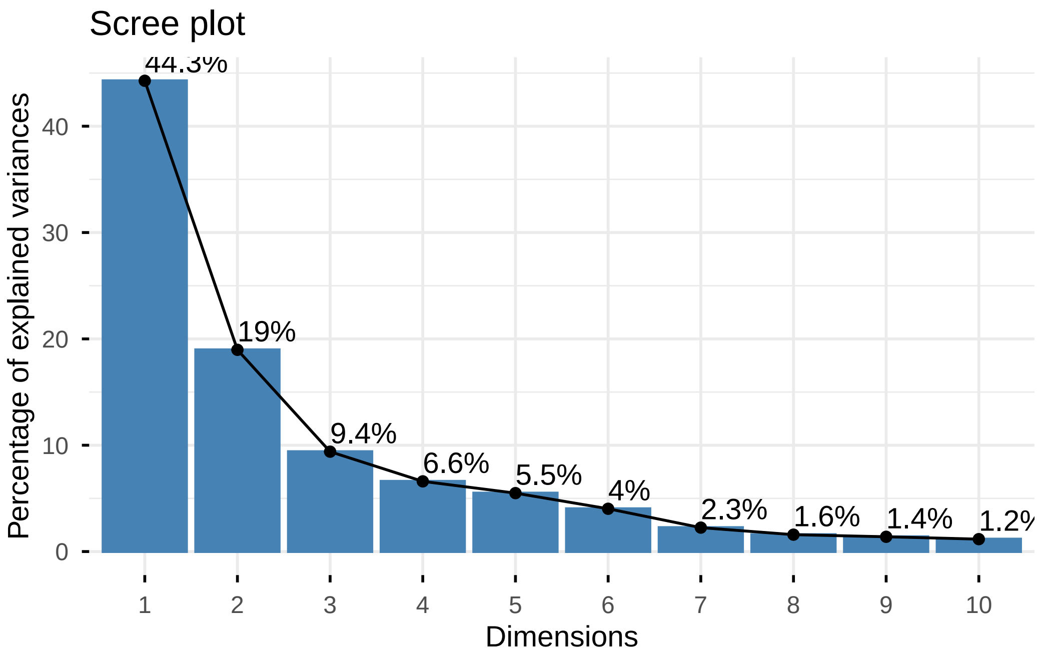

```
## ggplot based graph
#install.packages("factoextra")
library(factoextra)
```

```
Warning: package 'factoextra' was built under R version 4.5.2
```

```
Welcome to factoextra!
```

```
Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/
guide-to-principal-component-methods-in-r/
```

```
fviz_eig(wisc.pr, addlabels = TRUE)
```



Communicating PCA results

Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

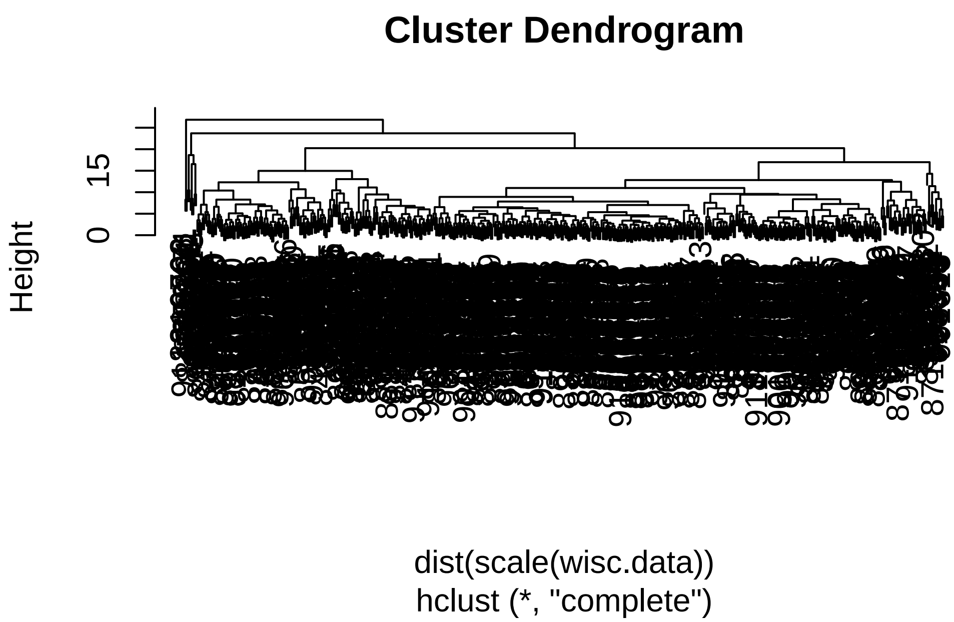
```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

The loading of concave.points_mean on PC1 is -0.2609, which indicates a strong contribution to the first principal component. There are other features with larger contributions, such as radius_mean.

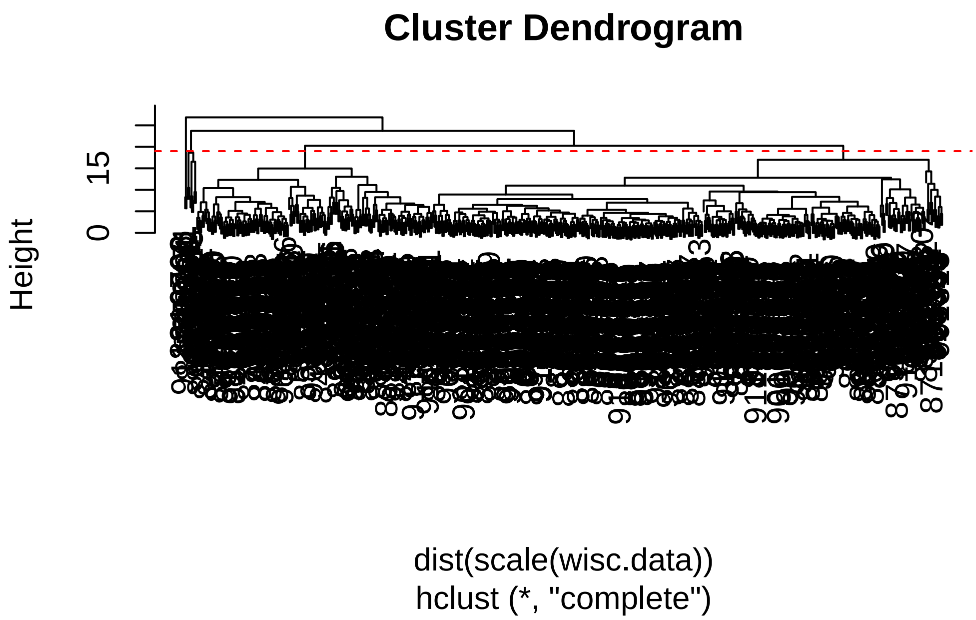
Hierarchical clustering

```
# Scale the wisc.data data using the "scale()" function
# Next, calculate the (Euclidean) distances between all pairs of observations in the new scal
wisc.hclust <- hclust(dist(scale(wisc.data)))
plot(wisc.hclust)
```



Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclust)
abline(h = 19, col = "red", lty = 2)
```




```r
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
table(wisc.hclust.clusters, diagnosis)
```

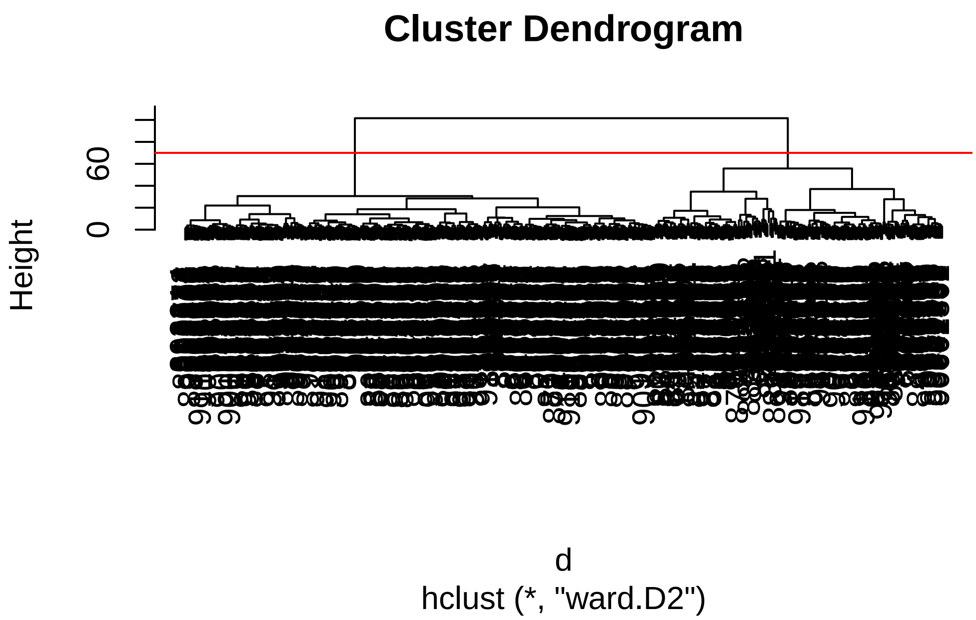
```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

> Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

The ward.D2 method gives the best results for this dataset. It produces more clearly separated clusters compared to methods like "single", "complete", and "average". Because ward.D2 minimizes variance within clusters, it forms spherical clusters, which aligns well with the structure of this data and results in better separation of malignant and benign samples.

Combining methods

```r
d <- dist(wisc.pr$x[,1:7])
wisc.pr.hclust <- hclust(d, method = "ward.D2")
plot(wisc.pr.hclust)
abline(h = 70, col = "red")
```




```r
grps <- cutree(wisc.pr.hclust, k = 2)
table(grps)
```

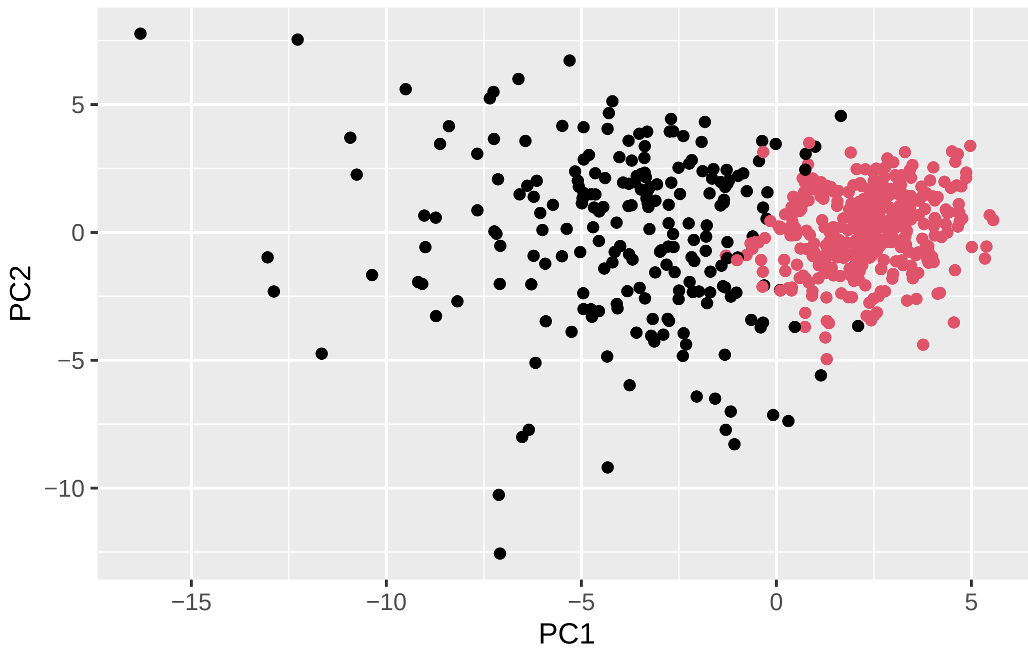
```
grps
  1   2
216 353
```

How does this clustering `grps` correspond to the expert `diagnosis`?

```r
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  28 188
   2 329  24
```

```r
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```




```r
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
```

> Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

```r
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

The newly created hclust model with two clusters separate out the two diagnoses pretty well with only 52 miscalculations. This shows an approximately 91% accuracy.

> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

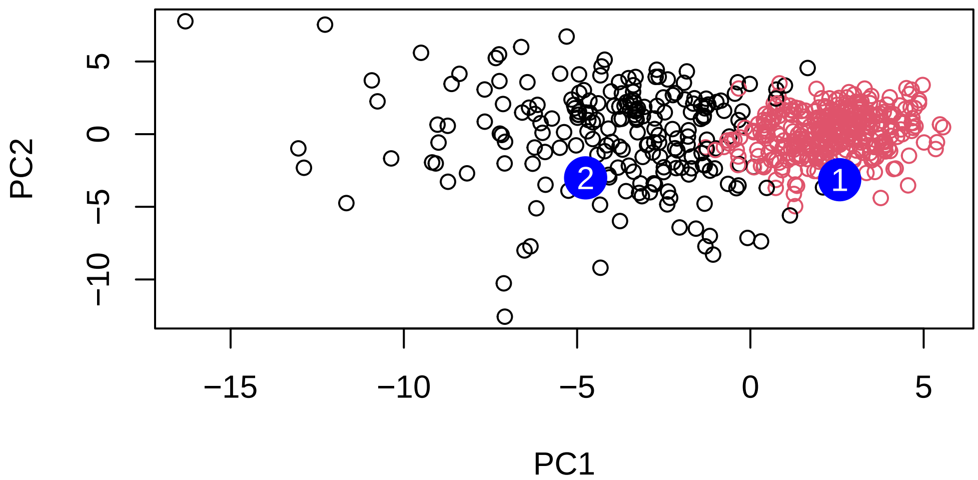
The hierarchical clustering model without PCA performs worse at separating malignant and benign samples. The clusters are more fragmented and show greater mixing of diagnoses.

Prediction

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



> Q16. Which of these new patients should we prioritize for follow up based on your results?

Patient 1 should be prioritized for follow up based on these results, as their PCA projection is closer to the malignant cluster, while Patient 2 appears more consistent with benign samples.

About this document

```r
sessionInfo()
```

```
R version 4.5.1 (2025-06-13)
Platform: x86_64-apple-darwin20
Running under: macOS Sequoia 15.4.1

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-x86_64/Resources/lib/libRlapack.dylib;

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
```

```
time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] factoextra_2.0.0 ggplot2_4.0.2

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       jsonlite_2.0.0     dplyr_1.2.0        compiler_4.5.1
 [5] ggsignif_0.6.4     tidyselect_1.2.1   Rcpp_1.1.0         tidyr_1.3.1
 [9] scales_1.4.0       yaml_2.3.10        fastmap_1.2.0      R6_2.6.1
[13] ggpubr_0.6.3       labeling_0.4.3     generics_0.1.4     Formula_1.2-5
[17] knitr_1.50         backports_1.5.0    ggrepel_0.9.6      tibble_3.3.0
[21] car_3.1-3          pillar_1.11.1      RColorBrewer_1.1-3 rlang_1.1.7
[25] broom_1.0.10       xfun_0.53          S7_0.2.0           cli_3.6.5
[29] withr_3.0.2        magrittr_2.0.4     digest_0.6.37      grid_4.5.1
[33] rstudioapi_0.17.1  lifecycle_1.0.5    vctrs_0.7.1        rstatix_0.7.3
[37] evaluate_1.0.5     glue_1.8.0         farver_2.1.2       abind_1.4-8
[41] carData_3.0-5      rmarkdown_2.30     purrr_1.1.0        tools_4.5.1
[45] pkgconfig_2.0.3    htmltools_0.5.8.1
```